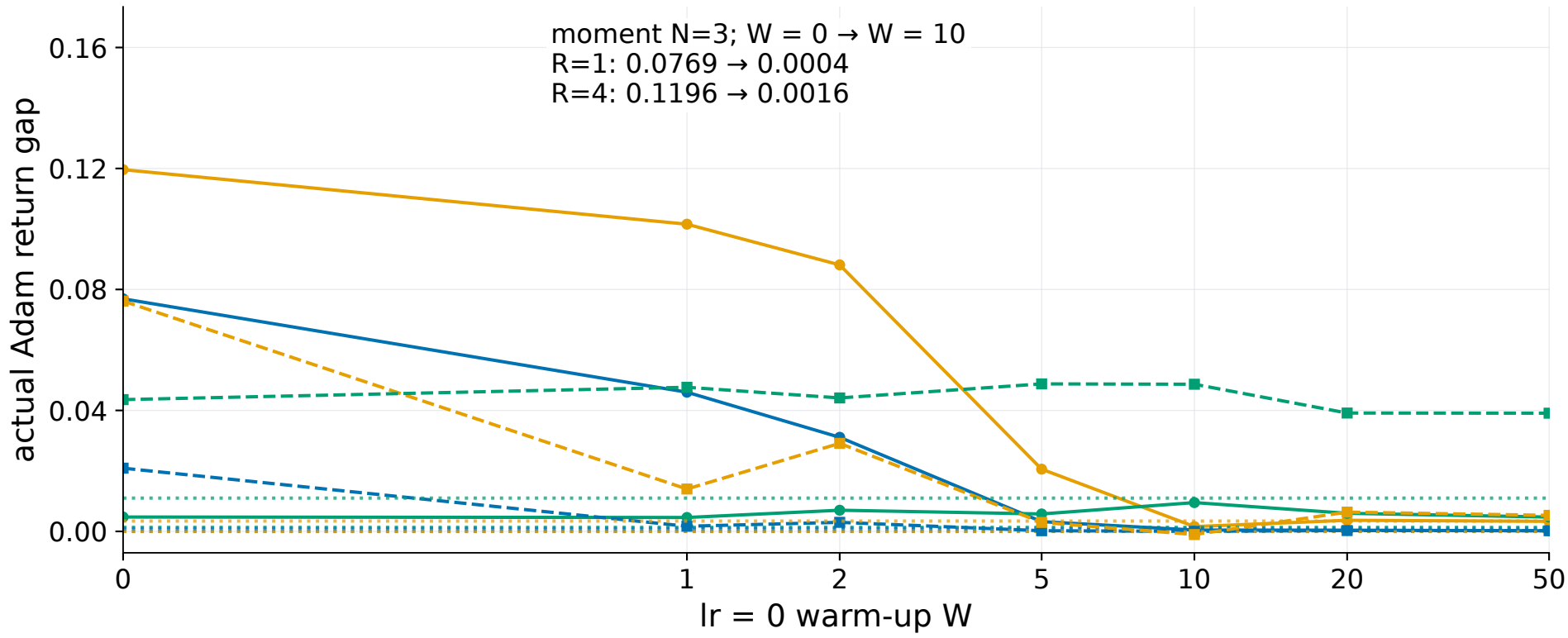
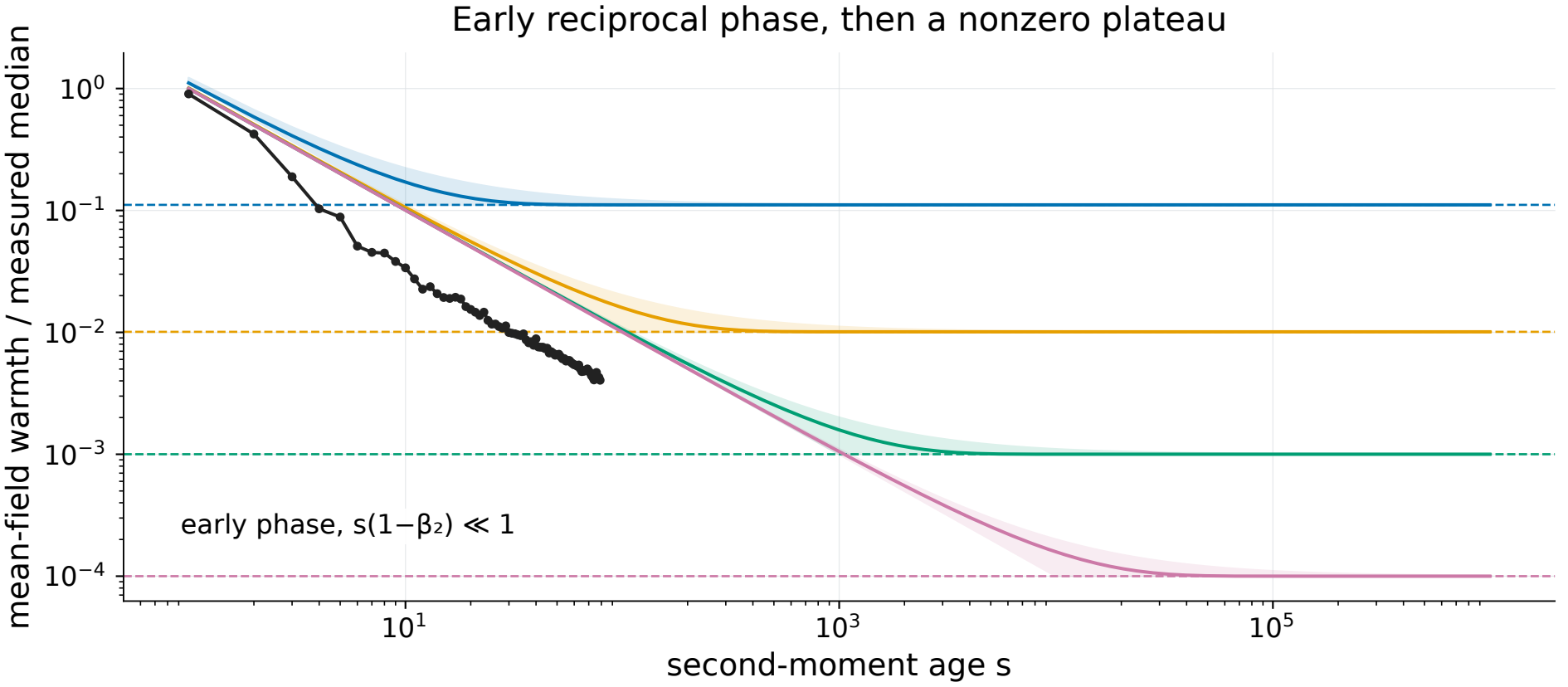
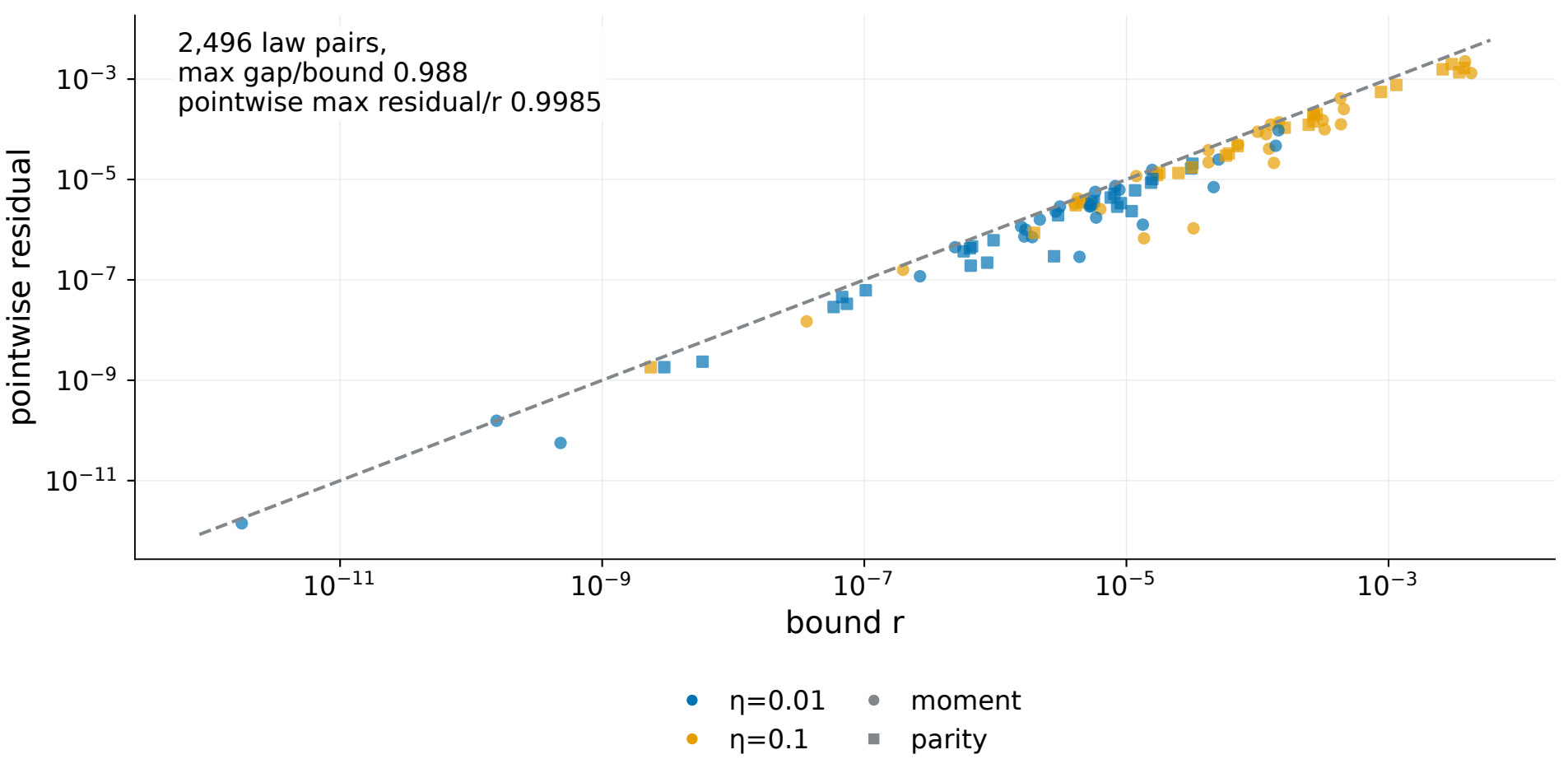
Gap vs lr=0 warm-up W



Early reciprocal phase, then a nonzero plateau



Warm-Adam affine residual vs bound



Controlled small network, not a real-model result; the Prop. 5.6 check is empirical.